

Chandra Prakash Bathula

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EDUCATION

Saint Louis University, School for Professional Studies <i>M.S. in Information Systems, GPA: 3.93/4.00 Awarded Distinction for Research Excellence</i>	May 2024 St. Louis, MO
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TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL, R
ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, Transformers, BERT, CNNs, ResNet, LSTM, GRU, XGBoost
GenAI: HuggingFace, OpenAI API, Gemini API, LangChain, RAG, Agentic RAG, Supabase pgvector, Prompt Engineering
Backend/Infra: FastAPI, PostgreSQL, Docker, AWS (S3, EC2, RDS), GCP (Vertex AI, BigQuery), Firebase, Git, CI/CD
Frontend: React.js, Next.js, Vue.js, Node.js, Redux, Responsive Design, WCAG/ARIA
Testing/Tools: Jest, Cypress, GitHub Actions, Webpack, Parcel, Figma, Unit/Integration/E2E Testing

EXPERIENCE

Saint Louis University — School for Professional Studies <i>AI/ML Engineer & Researcher Adjunct Faculty (IS 5200, IS 5500, IS 5600, AA 5750)</i>	Aug 2024 – Present St. Louis, MO
- Architected production-grade multi-agent RAG pipelines using Supabase pgvector for semantic search, HuggingFace Transformers for inference, LangChain for agent orchestration, and OpenAI APIs for reasoning; integrated automated evaluation frameworks to measure retrieval precision and response quality, designed for horizontal scalability, and raised assignment completion from 60% to 95% across three cohorts, improving course success rates 35%. - Led technical direction for 300+ engineers across 20+ ML capstone projects spanning full-lifecycle data pipelines: ingestion, labeling, model training, optimization, evaluation, and cloud deployment; reduced dataset-induced training errors 25%. - Built reusable Docker-containerized lab environments with GitHub Actions CI/CD and automated grading pipelines backed by PostgreSQL; cut instructor setup time 50% and achieved 100% uptime for student environments.	
Saint Louis University — iPaaS & Internal Tools Team <i>Software Engineer Intern, AI & Workflow Automation</i>	Mar 2023 – Dec 2023 St. Louis, MO
- Built a Python ML-based priority routing engine for SLU's internal tools platform; reduced manual triage 40% and improved SLA compliance 30% across 5K+ monthly service requests. - Trained and deployed a CNN/ResNet-50 image classification system for Lost-and-Found automation using PyTorch; reduced manual sorting time 35% and improved item matching accuracy 28% over 10K+ monthly photographs. - Deployed ML microservices with Docker and cloud compute auto-scaling; engineered FastAPI REST endpoints with Redis caching, reducing CNN inference latency from 2.5s to 250ms under 100+ concurrent requests.	
Qentelli Solutions Pvt Ltd <i>Associate Software Engineer</i>	Mar 2021 – Jul 2022 Hyderabad, India
- Built production front-end features for 3 enterprise web applications serving 5K+ users using TypeScript, React.js, Vue.js, and Redux; translated Figma designs into pixel-perfect, mobile-first, WCAG-accessible components. - Configured Webpack 5 with tree-shaking and code-splitting; enforced strict TypeScript + ESLint standards and expanded Jest coverage to 92%, improving release deployment efficiency 25%. - Integrated LSTM/GRU, Random Forest, and XGBoost models into automated QA pipelines using scikit-learn; reduced defect false positives 18% and improved anomaly detection precision 22% over 50K+ monthly test executions.	

PROJECTS

Agentic RAG Automation <i>Gmail, Google Drive, WhatsApp, OpenAI, LangChain, pgvector</i>
Built a production multi-agent RAG system using Gmail and Google Drive as private vector knowledge sources; automated context-aware email drafting and triggered WhatsApp alerts for unresolved queries with agentic orchestration.
Research Retrieval Agent <i>ArXiv, DuckDuckGo, OpenAI Function Calling, LangChain</i>
Autonomous multi-agent system that retrieves 5+ research papers, extracts structured metadata, summarizes findings, and generates insights using OpenAI function calling end-to-end agentic pipeline design.
Food-101 Image Classifier <i>ResNet-50, PyTorch, HuggingFace Hub, Gradio, torchvision</i>
Fine-tuned ResNet-50 (ImageNet pretrained) on 101K images across 101 classes for 100 epochs, surpassing 80% accuracy; deployed live Gradio inference demo on HuggingFace Spaces full ML lifecycle from training to production.

PUBLICATIONS & ACHIEVEMENTS

Author, ACM 2025: “Developing an Enterprise Application Tool to Discover Midwest Job Trends” — <i>Journal of Computing Sciences in Colleges</i> , 40(6), pp. 75–89. doi:10.5555/3729857.3729866
Awarded Distinction & Distinguished Graduate for MRP Thesis and overall GPA — M.S. in Information Systems, Saint Louis University, 2024.